

Qilin Huang

Pittsburgh, PA | qilinhua@andrew.cmu.edu

Seeking Summer 2027 Internships

EDUCATION

Carnegie Mellon University

Master of Science in Computer Vision

Current Coursework: Advanced Computer Vision; Visual Learning and Recognition; Introduction to Robot Learning.

Pittsburgh, PA

Expected Dec. 2027

Southern University of Science and Technology

Bachelor of Engineering in Computer Science and Technology

GPA: 3.89/4.00 | **Rank:** 13/167

Selected Coursework: Natural Language Processing; Intelligent Robotics.

Shenzhen, China

Jun. 2026

TECHNICAL SKILLS

Programming: Python, Java, C/C++, SQL.

Libraries & Tools: PyTorch, NumPy, scikit-learn, Linux, SLURM.

RESEARCH & PROJECTS

UniPixie: Controllable 3D Physics

Research Project | GRASP Laboratory, University of Pennsylvania

CVPR 2026 Highlight

Jul. 2025 - Mar. 2026

Philadelphia, PA / Remote

- Built a **conditional flow-matching model** to infer spatially varying material properties from visual features, with scalar conditioning for **continuous stiffness control**.
- Designed comparative experiments and ablation studies; UniPixie achieved **55.6% lower** MSE in log Young's modulus than retrained PIXIE on PixieMultiVerse.
- Developed unified interfaces connecting model predictions to **three existing simulation backends**: Material Point Method, linear blend skinning, and spring-mass systems.

Neural Module Network for 3D Visual Question Answering

Independent Research Project | Southern University of Science and Technology

- Proposed and implemented a 3D VQA prototype using a frozen LLM to translate natural-language questions into executable programs.

- Integrated task-specific 3D modules for step-by-step reasoning and evaluated the approach on ScanQA.

Feb. - Jun. 2025

Shenzhen, China

HCNQA: 3D Visual Question Answering

Research Project | Southern University of Science and Technology

- Reproduced baseline experiments on ScanQA for comparative evaluation and authored the VQA related-work section.

Jun. - Sep. 2024

Shenzhen, China

Face Recognition under Occlusion

Visual Computing Course Project | National University of Singapore

- Implemented face alignment, cropping, and feature extraction; compared preprocessing and feature methods for masked and unmasked face recognition.

May - Jul. 2024

Singapore

PUBLICATIONS

UniPixie: Unified and Probabilistic 3D Physics Learning via Flow Matching

Qilin Huang et al. **CVPR 2026, Highlight.** Co-first author.

HCNQA: Enhancing 3D VQA with Hierarchical Concentration Narrowing Supervision

Shengli Zhou, Jianuo Zhu, **Qilin Huang**, et al. **ICANN 2025.**